MIGRATION PLAN & EXECUTION GUIDE

J-Vo System Migration

Oracle Forms 6i / Oracle 8 → .NET Application / SQL Server

Enterprise Legacy System Modernization | Full ETL Lifecycle | Zero-Downtime Cutover Strategy

DOCUMENT PURPOSE

This document outlines the complete end-to-end migration strategy and execution plan for transitioning the J-Vo system from a legacy Oracle Forms 6i front-end with Oracle 8 database to a modern .NET application backed by Microsoft SQL Server. It covers all migration lifecycle phases: Project Initiation, Environment Setup, Extract, Data Profiling, Transform, Table Structure & Key Mapping, Clean, Load (Dry Runs), Security & Access Control, Performance Testing, Training, Change Management, UAT, Parallel Run, Cutover, Hypercare, and Legacy Decommission.

Document Info	Details	Status
Project	J-Vo System Migration	
Source System	Oracle Forms 6i + Oracle 8 Database	
Target System	.NET Application + Microsoft SQL Server	
Document Type	Migration Plan & Execution Guide	
Classification	Portfolio Sample : Demonstrates Migration Expertise	

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1. Executive Summary

The J-Vo system migration is a strategic initiative to retire a legacy Oracle Forms 6i application and its Oracle 8 database and replace it with a modern, maintainable .NET web application using Microsoft SQL Server as the data platform. This initiative addresses critical risks associated with the aging technology stack including: end-of-support for Oracle Forms 6i, limited developer availability for legacy Oracle technologies, inability to integrate with modern APIs and third-party services, and increasing infrastructure maintenance overhead.

This document serves as the authoritative migration plan, detailing each phase of the data and application transition lifecycle from initial extraction through production go-live, and encompasses all technical, functional, and organizational activities required for a successful cutover.

Key Migration Objectives

Business Objective	Technical Objective
Eliminate Oracle license and support costs	Migrate all schema objects and data to SQL Server
Modernize user experience with .NET UI	Re-engineer data types, sequences, and constraints
Improve system performance and scalability	Validate data integrity via reconciliation reports
Enable cloud-readiness and API integrations	Achieve zero data loss through validated dry runs
Reduce time-to-market for new features	Complete UAT sign-off before production cutover

2. Migration Overview & Architecture

SOURCE	Oracle Forms 6i (Front-End) | Oracle 8i Database | PL/SQL Business Logic | Oracle Reports
TARGET	.NET Application (ASP.NET / C#) | Microsoft SQL Server | T-SQL Stored Procedures | SSRS / .NET Reports

Migration Approach

The migration follows a phased ETL (Extract, Transform, Load) methodology combined with structured testing gates and organizational change management activities. The approach prioritizes data fidelity above all else, using multiple dry-run load cycles to validate completeness and accuracy before the final production cutover is executed.

High-Level Phase Timeline

Phase	Activity	Key Deliverable	Gate
0	**Project Initiation & Discovery**	Project charter, scope doc, environment inventory	Sponsor sign-off
0B	**Environment & Infrastructure Setup**	Dev / Staging / UAT / Prod environments ready	Infra readiness confirmed
1	**Extract**	Data export files, schema documentation	Data completeness sign-off
1B	**Data Profiling**	Data profile report, anomaly inventory	Profiling report approved
2	**Transform**	Mapped data sets, transformation scripts	Mapping approval
2A	**Table Structure & Key Mapping**	PK/FK mapping doc, key validation scripts	DBA + Lead sign-off
3	**Clean**	Cleansed data sets, data quality report	Quality threshold met
4	**Load / Dry Runs**	Validated SQL Server database (3 dry runs)	Dry Run 3 pass
4B	**Security & Access Control**	Role mapping doc, user accounts provisioned in SQL Server	Security review passed
4C	**Performance & Non-Functional Testing**	Performance test results, SLA sign-off	SLA thresholds met
5	**Training**	Trained end users, training materials	Training completion sign-off

5B	**Change Management & Comms**	Stakeholder comms sent, change plan executed	Comms plan complete
6	**UAT**	UAT sign-off document, defect closure report	UAT pass / go-live approval
6B	**Parallel Run**	Side-by-side output comparison report	Business parity confirmed
7	**Cutover & Go-Live**	Production system live, rollback tested	Go/No-Go decision passed
8	**Hypercare / Stabilization**	Issue log, daily stand-ups, stabilization sign-off	Hypercare exit sign-off
9	**Legacy Decommission**	Oracle system retired, licenses released, data archived	Decommission sign-off

PHASE 00

Project Initiation & Discovery

Scope Definition, Stakeholder Alignment & Migration Strategy Selection

0.1 Objectives

Before any technical work begins, the migration must be formally chartered, scoped, and resourced. The Project Initiation phase establishes the boundaries of the migration, identifies all stakeholders, inventories the source and target environments, and selects the migration strategy. Skipping or rushing this phase is one of the most common root causes of failed migrations.

0.2 Initiation Activities

- Develop and obtain sign-off on the Project Charter : defines scope, budget, timeline, sponsor, and success criteria
- Conduct a Source System Inventory: catalog all Oracle Forms 6i screens, Oracle 8 schemas, PL/SQL objects, Oracle Reports, interfaces, and integrations
- Conduct a Target System Inventory: document the .NET application architecture, SQL Server version, hosting environment, and deployment model
- Identify and document all data consumers: batch jobs, downstream systems, APIs, third-party integrations that read from or write to Oracle 8
- Define migration scope boundaries: what data migrates, what is archived, what is discarded : approved by Business Owner
- Select migration strategy: Big Bang (single cutover) vs. Phased (module-by-module) vs. Parallel Run : document decision rationale
- Establish the project team: assign all roles defined in Section 10 with named individuals and backups
- Stand up the project governance model: weekly status meetings, escalation path, decision log, and change control process
- Create and baseline the project schedule with all phase milestones and gate dates
- Establish the document repository and version control baseline for all migration artifacts

0.3 Migration Strategy Decision

Strategy	Description	Best Fit When
Big Bang Cutover	All data and users migrate in a single maintenance window; legacy system turned off at cutover	System is self-contained; cutover window is feasible; business can tolerate brief downtime
Phased Migration	Modules or user groups migrate incrementally over multiple releases; legacy and new run simultaneously per module	Large system; cannot tolerate full downtime; complex integrations require staged validation
Parallel Run	Both systems run simultaneously; outputs are compared before legacy is decommissioned	High-risk data; regulatory requirements; business needs confidence period before full switchover

J-Vo STRATEGY

The J-Vo migration employs a Big Bang Cutover with a mandatory Parallel Run validation window prior to final decommission. This balances speed of transition with confidence in data and output parity.

0.4 Exit Criteria

- Project Charter signed by Project Sponsor
- Source and target system inventory documents complete and baselined
- Migration scope document approved by Business Owner
- All project team roles filled with named individuals
- Project schedule baselined with all milestone dates agreed
- Document repository established and all team members have access

PHASE 0B

Environment & Infrastructure Setup

SQL Server, .NET, Tooling & Network Readiness

0B.1 Objectives

The migration cannot proceed without a properly configured set of non-production environments that mirror the production architecture. This phase establishes the Development, Staging, UAT, and Production environments for SQL Server and the .NET application, and ensures all migration tooling, network connectivity, and access controls are in place before data work begins.

0B.2 Required Environments

Environment	Purpose	Data	Access
Development	ETL script development, transformation unit testing, schema creation iterations	Subset of Oracle extract : anonymized	Migration team + DBA
Staging / Migration Lab	Full dry run loads, reconciliation testing, performance benchmarking	Full Oracle extract : anonymized	Migration team + DBA + Dev team
UAT	User acceptance testing, report validation, business sign-off	Dry Run 3 load : full representative data	UAT testers + Support + Dev team
Training	End-user training sessions, Super User workshops	Sanitized representative data : not production	Training Coordinator + End users
Production	Live system post-cutover	Production load during cutover window	Authorized users only

0B.3 Infrastructure & Tooling Checklist

- SQL Server instances provisioned for all environments at the correct version and edition
- SQL Server Agent configured for scheduled jobs; SQL Server Integration Services (SSIS) installed if used for ETL
- .NET application deployed to each environment from the correct release branch
- Network connectivity confirmed: migration lab server can reach Oracle 8 source server (read-only) and SQL Server target
- Oracle client tools installed on migration workstations: SQL*Plus, Data Pump client, Oracle ODBC/OLE DB drivers
- SSMS (SQL Server Management Studio) and any approved ETL tooling installed and licensed
- Version control repository configured for all migration scripts : no manual ad-hoc scripts executed outside version control
- Backup jobs configured and tested on all SQL Server environments before any data is loaded
- Monitoring and alerting configured on production SQL Server (disk space, CPU, memory, blocking queries)

- VPN or secure tunnel established if migration environments span different network zones
- Service accounts and connection strings documented in the secure configuration store : no hardcoded credentials

0B.4 Exit Criteria

- All five environments provisioned and verified by the DBA
- Network connectivity tests pass between all components
- Backup and restore procedure verified on staging environment
- All migration tooling installed and team access confirmed
- Infrastructure Readiness sign-off issued by DBA and Infrastructure Lead

PHASE 01

Extract

Oracle 8 Data Extraction & Source System Documentation

1.1 Objectives

The Extract phase captures a complete, point-in-time snapshot of all data residing in the Oracle 8 database. This phase also documents the source schema in full so the transformation team has an authoritative reference during mapping activities.

1.2 Activities

- Inventory all Oracle 8 schemas, tables, views, sequences, stored procedures, triggers, and packages
- Identify all data volumes per table and document row counts, nullable fields, and primary/foreign key relationships
- Execute full data export using Oracle Data Pump (expdp) or SQL*Plus spool scripts for legacy Oracle 8 compatibility
- Export BLOB/CLOB objects separately with appropriate handling scripts
- Document all Oracle-specific data types: NUMBER, VARCHAR2, DATE, CLOB, BLOB, ROWID, RAW
- Capture all PL/SQL stored procedures and triggers for logic re-implementation review
- Snapshot application configuration tables and code/lookup tables
- Extract all Oracle Reports (.rdf) files and document report data sources
- Generate extraction manifest listing every exported object, file name, and checksum

1.3 Technical Approach : Oracle 8 Export Commands

Method	Usage
exp utility (Oracle 8)	exp userid=system/pwd file=jvo_full.dmp full=y log=export.log
SQL*Plus spool	spool table_data.csv : for selective table exports
PL/SQL script	Custom cursor loops for tables with LOB columns
Schema DDL	dbms_metadata.get_ddl() for all objects

1.4 Exit Criteria

- All tables exported with row counts reconciled to production totals
- Extraction manifest approved by DBA and Project Manager
- Source schema documented and baselined under version control
- All files transferred to migration working environment and checksums verified

PHASE 1B

Data Profiling

Source Data Analysis Before Transformation Design

1B.1 Objectives

Data Profiling is a discrete analytical phase that occurs after extraction and before transformation design is finalized. While Extract captures the data, Profiling interrogates it : revealing the actual content, patterns, anomalies, and quality characteristics of every column and relationship. Profiling findings directly feed into the Data Mapping Specification and the Clean phase scope. Attempting to write transformation scripts without profiling leads to late-discovered surprises during dry runs.

1B.2 Profiling Dimensions

Dimension	What Is Measured	Migration Impact
Completeness	NULL rate, blank rate, and missing value rate per column	Identifies fields needing default values or exclusion rules in Transform
Uniqueness	Duplicate row rate; count of distinct values vs. total rows	Identifies de-duplication need; confirms PK candidate viability
Validity	Values outside domain, format violations, constraint violations	Scopes the Clean phase; informs lookup/code table mapping
Consistency	Cross-column and cross-table value agreement (e.g., status vs. date fields)	Reveals business rule violations that survive to migration data
Timeliness	Distribution of data by date range; identification of stale or future-dated records	Informs archival scope decision; flags records for business review
Referential Integrity	Actual FK orphan counts across all parent-child relationships	Directly scopes orphan remediation work in Clean phase
Data Volume	Row counts, table sizes in MB/GB, LOB sizes	Drives load window estimation and infrastructure sizing
Value Distribution	Frequency histograms of key categorical columns (status, type codes)	Validates lookup table completeness; reveals unexpected code values

1B.3 Profiling Outputs

- Data Profile Report : one section per table; covers all eight profiling dimensions
- Anomaly Inventory : prioritized list of data issues by severity and volume, with recommended disposition
- Transformation Impact List : profiling findings that require changes to the Data Mapping Specification
- Clean Phase Scope Document : input to Phase 3; defines exactly which issues must be remediated and which are acceptable

1B.4 Exit Criteria

- Data Profile Report completed for all tables in scope
- Anomaly Inventory reviewed and dispositions assigned by Business Owner and BA
- Transformation Impact List incorporated into the Data Mapping Specification
- Clean Phase Scope Document issued and approved

PHASE 02

Transform

Data Mapping, Type Conversion & Business Logic Re-Engineering

2.1 Objectives

The Transform phase converts extracted Oracle data into a format compatible with the SQL Server target schema. This includes data type mapping, re-engineering of PL/SQL business logic to T-SQL, and translation of Oracle-specific constructs to their SQL Server equivalents.

2.2 Data Type Mapping : Oracle to SQL Server

Oracle 8 Type	SQL Server Type	Notes	Action Required
NUMBER(p,s)	DECIMAL(p,s)	Direct mapping	Verify precision
NUMBER(p,0)	INT / BIGINT	Integer types	Check range
VARCHAR2(n)	NVARCHAR(n)	Unicode support added	Encoding review
CHAR(n)	CHAR(n)	Direct mapping	Pad/trim check
DATE	DATETIME2	Oracle DATE includes time	Time component audit
CLOB	NVARCHAR(MAX)	Large text content	Size validation
BLOB	VARBINARY(MAX)	Binary objects	Storage path review
ROWID	UNIQUEIDENTIFIER	Row identifiers	Regenerate in SQL Server
SEQUENCE	IDENTITY / SEQUENCE	Auto-increment logic	Seed value alignment
BOOLEAN (PL/SQL)	BIT	Oracle has no SQL BOOLEAN	0/1 conversion

2.3 Transformation Activities

- Develop column-level data mapping document (Source Table.Column → Target Table.Column)
- Write SSIS packages or Python/T-SQL transformation scripts for each entity group
- Convert Oracle sequences to SQL Server IDENTITY columns or SEQUENCE objects with matching seed values
- Re-implement PL/SQL stored procedures as T-SQL stored procedures in SQL Server
- Convert Oracle triggers to SQL Server triggers or .NET application-layer logic as appropriate
- Translate Oracle NVL(), DECODE(), ROWNUM to their T-SQL equivalents (ISNULL, CASE, ROW_NUMBER())
- Address Oracle DATE to DATETIME2 conversion, paying particular attention to implicit time components
- Map Oracle schema owner prefixes to SQL Server database schemas
- Document all transformation decisions in the Data Mapping Specification document

2.4 Exit Criteria

- Data Mapping Specification reviewed and approved by Business Analyst and DBA
- All transformation scripts unit-tested against sample data sets
- PL/SQL to T-SQL conversion reviewed by SQL Server DBA
- No unresolved mapping items : all exceptions documented with disposition decisions

2A. Table Structure Divergence & Key Mapping

A critical risk factor in the J-Vo migration is that the Oracle 8 source schema and the SQL Server target schema are structurally different : they are not a direct table-for-table, column-for-column replica. Table names, column names, data types, primary key strategies, and foreign key relationships have all been redesigned in the new .NET application. This section documents the categories of structural difference, the key mapping strategy, and the validation controls required to ensure no data is lost or misrouted during transformation and load.

WARNING Do NOT assume any Oracle table maps directly to a SQL Server table of the same name. Every table mapping must be explicitly documented in the Data Mapping Specification before transformation scripts are written.

2A.1 Categories of Table Structure Differences

Difference Category	Oracle 8 Pattern	SQL Server / .NET Pattern
Table Naming Convention	ALL_CAPS with underscores (e.g., CUSTOMER_MASTER)	PascalCase or mixed (e.g., CustomerMaster or tbl_CustomerMaster)
Column Naming Convention	ALL_CAPS abbreviated (e.g., CUST_ID, CUST_NM)	Descriptive camelCase or PascalCase (e.g., CustomerId, CustomerName)
Table Consolidation	Multiple narrow tables for a single entity	Denormalized into fewer, wider tables in the new schema
Table Splitting	Single wide table handling multiple entity types	Decomposed into separate normalized tables per entity
Junction / Bridge Tables	Implicit relationships via composite keys	Explicit junction tables with surrogate keys
Audit Columns	No standard audit columns in legacy schema	CreatedDate, ModifiedDate, CreatedBy added to all tables
Soft Delete Pattern	Active flag column (ACTIVE_FL = Y/N)	IsDeleted BIT column with NULL = active convention
Schema Owner Prefix	Oracle schema owner prefix (e.g., JVOAPP.ORDERS)	SQL Server database schema prefix (e.g., dbo.Orders)
Computed Columns	Calculated via PL/SQL function or view	Persisted computed columns or application-layer calculation

2A.2 Primary Key Strategy Differences

The primary key strategy between Oracle 8 and SQL Server differs fundamentally. Oracle relied on SEQUENCE objects and NUMBER data types to generate surrogate keys, often through BEFORE INSERT triggers. The new SQL Server schema uses IDENTITY columns and, in select cases, UNIQUEIDENTIFIER (GUID) columns for distributed key generation. This difference requires explicit handling during transformation to prevent key collisions and broken relationships.

Aspect	Oracle 8 Approach	SQL Server Approach
PK Generation Mechanism	Oracle SEQUENCE + BEFORE INSERT trigger assigns next value to NUMBER PK column	IDENTITY(seed, increment) on INT or BIGINT column : auto-assigned on insert
PK Data Type	NUMBER(10) or NUMBER(15) : no true integer type	INT (up to 2.1B rows) or BIGINT : explicit integer; select tables use UNIQUEIDENTIFIER
PK Column Naming	Typically <TABLE_ABBREV>_ID (e.g., CUST_ID, ORD_ID)	Typically <EntityName>Id using PascalCase (e.g., CustomerId, OrderId)
Composite Primary Keys	Widely used : multiple columns form the PK	Minimized in new schema; surrogate single-column PKs preferred with UNIQUE constraints for natural keys
Sequence Seed Alignment	Sequences may have large gaps from deletes or reseeds	IDENTITY seed set to max(legacy_pk_value) + 1000 to prevent collisions with migrated data
GUID / UUID Usage	Not present in Oracle 8 schema	Select tables (e.g., integration-facing entities) use NEWSEQUENTIALID() for UNIQUEIDENTIFIER PKs

PK ACTION	For every table: (1) Migrate Oracle NUMBER PK values as-is into SQL Server target columns during load : do not allow IDENTITY to auto-generate during migration. (2) Use SET IDENTITY_INSERT <table> ON during load scripts. (3) After load completes, reseed all IDENTITY columns to max(PK) + 1000 using DBCC CHECKIDENT. (4) Validate no sequence gaps will cause FK violations.

2A.3 Foreign Key Strategy Differences

Foreign key relationships in the Oracle 8 schema used NUMBER-to-NUMBER references with naming conventions that do not carry over to SQL Server. In addition, the new schema introduces new FK relationships that did not exist in Oracle (due to normalization), and some Oracle FK relationships are consolidated or removed in the new design. All FK changes must be catalogued before the Clean phase begins.

Aspect	Oracle 8 Approach	SQL Server Approach
FK Data Type	NUMBER referencing a NUMBER PK : no explicit INT type	INT or BIGINT referencing an INT/BIGINT IDENTITY PK : strict type matching enforced
FK Naming Convention	FK_<CHILD_TABLE>_<PARENT_TABLE> pattern or no naming standard	FK_<ChildTable>_<ParentTable>_<Column> : enforced naming standard
Deferrable Constraints	Oracle supports DEFERRABLE INITIALLY DEFERRED FK constraints	SQL Server does not support deferrable constraints : load order dependency must be resolved explicitly
Cascading Rules	ON DELETE SET NULL or no action : often unenforced in Oracle 8	Explicit ON DELETE / ON UPDATE cascade rules defined per relationship in SQL Server

Self-Referencing FKs	Present (e.g., employee hierarchy, category tree)	Preserved : must be loaded in parent-before-child order; validate after load
New FKs in Target Schema	Relationships enforced only in PL/SQL application code	New FK constraints added to SQL Server schema that were not present in Oracle : orphan records will violate these on load
Removed FKs	Some Oracle FK constraints are consolidated or removed in redesign	Validate data consistency manually for relationships no longer enforced by a FK constraint

FK ACTION	During load: (1) Disable all FK constraints before bulk insert using ALTER TABLE ... NOCHECK CONSTRAINT ALL. (2) Load tables in strict parent-before-child dependency order. (3) Re-enable constraints using ALTER TABLE ... WITH CHECK CHECK CONSTRAINT ALL after all data is loaded. (4) Any constraint violation at re-enable is a blocking defect that must be resolved before proceeding to the next dry run.

2A.4 Sample Table Mapping : Oracle to SQL Server

The table below illustrates representative examples of how Oracle 8 tables map to SQL Server tables in the J-Vo migration. This is an excerpt; the complete mapping is maintained in the Data Mapping Specification document.

Oracle Table	Oracle PK	SQL Server Table	SQL Server PK	Map Type
CUSTOMER_MASTER	CUST_ID (NUMBER)	**dbo.Customer**	CustomerId (INT IDENTITY)	1-to-1
CUST_ADDRESS	ADDR_ID (NUMBER)	**dbo.CustomerAddress**	AddressId (INT IDENTITY)	1-to-1
ORDER_HDR	ORD_ID (NUMBER)	**dbo.Order**	OrderId (INT IDENTITY)	1-to-1
ORDER_DTL	ORD_DTL_ID (NUMBER)	**dbo.OrderLine**	OrderLineId (INT IDENTITY)	Renamed
PROD_MASTER	PROD_ID (NUMBER)	**dbo.Product**	ProductId (INT IDENTITY)	1-to-1
PROD_CATEGORY	CAT_ID (NUMBER)	**dbo.ProductCategory**	CategoryId (INT IDENTITY)	1-to-1
USR_MSTR + ROLES	USR_ID + ROLE_CD (Composite)	**dbo.AppUser + dbo.UserRole**	UserId (INT) + UserRoleId (INT IDENTITY)	Split
SYS_CODES	CODE_TYPE + CODE_VAL (Composite)	**dbo.LookupCode**	LookupCodeId (INT IDENTITY)	Consolidated
AUDIT_LOG	LOG_SEQ (NUMBER)	**dbo.AuditLog**	AuditLogId (BIGINT IDENTITY)	Type change

Map Types: 1-to-1 = direct table correspondence with column renames; Renamed = table renamed but structure similar; Split = one Oracle table becomes multiple SQL Server tables; Consolidated = multiple Oracle tables merged into one SQL Server table; Type change = PK data type changed.

2A.5 Key Validation Scripts : Required Per Dry Run

The following validation queries must be executed after every dry run load to confirm key integrity between the Oracle source and SQL Server target. All checks must return zero violations before the dry run is considered passed.

Validation Check	Purpose	Pass Threshold
PK uniqueness check	Confirm no duplicate PK values exist in any SQL Server target table after load	0 duplicates
PK NULL check	Confirm no NULL values exist in any PK column	0 NULLs
IDENTITY reseed verification	Confirm DBCC CHECKIDENT reports current identity value is greater than max loaded PK + 1000	All tables pass
FK constraint re-enable check	Confirm all FK constraints re-enabled WITH CHECK : no violations reported	0 violations
Orphan FK record scan	Query child tables for FK values with no matching parent PK : catches records missed by constraint check	0 orphans
PK value range continuity	Confirm migrated PK values fall within expected Oracle sequence range : detect any sequence misalignment	All in range
Cross-table row count reconciliation	Sum child record counts grouped by FK and compare to parent record counts for 1:many relationships	Counts reconcile
New FK constraint validation	Explicitly test FK constraints added in SQL Server that did not exist in Oracle : highest-risk violation points	0 violations

2A.6 Exit Criteria : Table Structure & Key Mapping

- Data Mapping Specification includes explicit Oracle Table.Column to SQL Server Table.Column mapping for every field
- All table-level mapping types (1-to-1, Split, Consolidated, Renamed) documented and approved
- PK migration strategy confirmed per table: IDENTITY reseed values calculated and scripted
- FK dependency load order documented and validated against actual schema constraints
- All new FK constraints in SQL Server schema identified and orphan-record pre-checks executed on Oracle source data
- Key validation scripts authored, reviewed, and incorporated into the dry run execution checklist
- DBA and Migration Lead sign off on complete key mapping document

PHASE 03

Clean

Data Quality Assessment, Remediation & Validation

3.1 Objectives

The Clean phase identifies and remediates data quality issues present in the legacy Oracle 8 data before it is loaded into the SQL Server target environment. Legacy systems often accumulate years of inconsistent, duplicate, orphaned, or otherwise non-conforming records that must be addressed before migration.

3.2 Data Quality Issue Categories

Issue Type	Description	Remediation Approach
Duplicate Records	Multiple rows representing the same entity	De-duplication scripts with survivor record selection rules
Orphaned Foreign Keys	Child records with no matching parent	Null-out or delete orphan records per business rules
NULL Violations	Required fields containing NULL values	Default value substitution or record exclusion
Invalid Codes/Lookups	Reference data values not in valid code tables	Map to closest valid code or flag for business review
Date Anomalies	Future dates, 01/01/1900 placeholders, impossible dates	Standardize or NULL with documentation
Character Encoding	Non-ASCII/extended characters from Oracle encoding	Transliterate or encode to Unicode
Trailing/Leading Spaces	Whitespace padding in VARCHAR/CHAR fields	TRIM() applied uniformly
Inconsistent Casing	Mixed case in name/code fields	Standardize to proper case or UPPER as required
Inactive/Archived Records	Records flagged inactive but referencing active data	Scope decision: migrate or exclude per business owner

3.3 Cleaning Activities

- Run automated data profiling against all extracted tables using SQL-based profiling scripts
- Generate Data Quality Assessment Report with issue counts by category and table
- Present findings to business stakeholders for disposition decisions on ambiguous records
- Implement cleansing scripts in a dedicated staging environment (do not modify source Oracle data)
- Re-profile cleansed data sets and produce before/after comparison report
- Document all records excluded from migration with business justification
- Validate referential integrity is intact across all foreign key relationships after cleansing

3.4 Data Quality Thresholds

THRESHOLD	Migration proceeds to Load phase only when: (a) Critical fields have 0% NULL violation rate, (b) Duplicate rate is < 0.1% of total records, (c) Referential integrity passes 100% on all FK relationships, (d) All business-owner sign-offs on disposition decisions are obtained.

3.5 Exit Criteria

- Data Quality Assessment Report reviewed and approved
- All cleansing scripts executed and validated in staging
- Referential integrity checks pass with 0 violations
- Business owner sign-off on all disposition decisions for edge-case records

PHASE 04

Load & Dry Runs

Iterative SQL Server Load Testing with Reconciliation Gates

4.1 Objectives

The Load phase populates the SQL Server target database with cleansed, transformed data. Rather than a single load, this phase prescribes a minimum of three successive Dry Runs. Each dry run is a full end-to-end load simulation allowing the team to identify and resolve technical, data, and performance issues before committing to the production cutover load.

4.2 Dry Run Structure

Run	Purpose	Success Criteria	Audience
Dry Run 1	Initial technical validation of load scripts, schema creation, and connectivity	Load completes with < 5% error rate; schema created without errors	Migration Team & DBA
Dry Run 2	Full data load with reconciliation; functional verification by the application team	Row counts within 0.5% of source; application connects successfully	Migration Team, DBA, Dev Team
Dry Run 3	Production-simulation load including timing; UAT environment ready	Row counts match 100%; load completes within cutover window; all app functions verified	All stakeholders
Production Load	Final go-live load executed during approved maintenance window	Zero data loss; application fully operational post-load	All stakeholders

4.3 Load Sequence : Per Dry Run

1. Restore clean SQL Server target database from baseline snapshot (truncate all tables)
2. Execute schema creation scripts (tables, indexes, constraints : constraints disabled for load)
3. Load data in dependency order: lookup/reference tables first, then master entities, then transactional data
4. Enable constraints and execute constraint validation scripts
5. Execute referential integrity validation queries
6. Run row count reconciliation: Oracle source counts vs. SQL Server target counts
7. Execute application smoke tests and functional checks
8. Capture load duration, error log, and reconciliation report
9. Conduct dry run retrospective and update defect log
10. Obtain dry run sign-off from DBA and Project Manager before proceeding to next run

4.4 Reconciliation Report : Required Metrics per Run

Metric	DR1 Target	DR2 Target	DR3 Target

Row count match (all tables)	> 95%	> 99.5%	100%
Constraint violations	< 100	< 10	0
FK integrity failures	< 50	0	0
NULL violations on required fields	< 200	< 5	0
Load duration (hours)	Baseline	Within 10% of DR1	Within cutover window
Application connectivity	Verified	Verified	Verified
Critical business function checks	N/A	Partial (50%)	100%

4.5 Exit Criteria

- Dry Run 3 passes all reconciliation thresholds at 100%
- Load duration confirmed to fit within the approved production maintenance window
- All critical application functions verified against SQL Server data
- DBA, Development Lead, and Project Manager provide written dry run sign-off
- Rollback procedure tested and documented

PHASE 4B

Security & Access Control Validation

User Provisioning, Role Mapping & Data Access Governance

4B.1 Objectives

Oracle Forms 6i managed user authentication and authorization through Oracle database users, roles, and application-level access logic embedded in PL/SQL. The new .NET application has a different identity model : typically using application-managed users stored in SQL Server, with role-based access control (RBAC) enforced by the .NET layer. This phase ensures that every Oracle user and permission is accurately translated to the new security model before go-live, and that no user gains excessive access as a result of a mis-mapped role.

4B.2 Security Migration Activities

- Extract and document all Oracle database users, their granted roles, and object-level privileges
- Extract and document all Oracle Forms function-level security assignments (which forms/menus each user or role can access)
- Map each Oracle role to its equivalent .NET application role : document any consolidations, splits, or new roles required
- Identify users whose Oracle access level does not have a direct equivalent in the new role model : escalate to Business Owner for decision
- Provision all user accounts in the SQL Server / .NET application environment with correct role assignments
- Validate that no privilege escalation has occurred: no user in the new system should have MORE access than they held in Oracle
- Configure SQL Server login accounts for application service accounts : principle of least privilege enforced
- Verify SQL Server Transparent Data Encryption (TDE) or equivalent encryption at rest is enabled if required by policy
- Confirm audit logging is active in SQL Server for all data modification operations
- Conduct a penetration/access review: attempt to access restricted data using a lower-privileged test account

4B.3 Oracle to .NET Role Mapping

Oracle Role / User Type	.NET Application Role	SQL Server Permission	Action Required
JVOAPP_ADMIN	System Administrator	db_owner on jvo_prod (via app svc account)	Verify : least privilege review
JVOAPP_POWERUSER	Power User / Manager	SELECT + INSERT + UPDATE on business tables	Map and test
JVOAPP_READONLY	Read-Only Viewer	SELECT only on reporting views	Map and test
JVOAPP_DATAENTRY	Standard User	SELECT + INSERT + UPDATE on assigned modules	Map and test

JVOAPP_REPORTS	Report Viewer	SELECT on reporting schema only	Map and test
Oracle DBA accounts	No application role : DB admin only	db_owner on SQL Server : DBA team only	Do NOT migrate to app users

4B.4 Exit Criteria

- All Oracle users mapped to .NET application roles : no unmapped users
- User accounts provisioned in SQL Server / .NET environment with correct role assignments
- Privilege escalation review complete : no user has more access than held in Oracle
- Service accounts configured with least-privilege permissions
- Security review sign-off from IT Security or designated security reviewer

PHASE 4C

Performance & Non-Functional Testing

Load Testing, SLA Validation & SQL Server Query Tuning

4C.1 Objectives

The .NET application and SQL Server must perform within defined Service Level Agreements (SLAs) under realistic production load conditions before go-live is authorized. Performance testing is executed against the SQL Server environment loaded with production-representative data from Dry Run 3. This phase surfaces missing indexes, inefficient queries, blocking conditions, and infrastructure sizing issues that would otherwise appear as post-go-live degradation.

4C.2 SLA Targets

Metric	Target SLA	Test Method
Standard page / screen load time	< 3 seconds at 50 concurrent users	Load test tool (JMeter / k6)
Complex search / query response	< 5 seconds at 50 concurrent users	Load test with representative queries
Report generation (standard)	< 30 seconds	Automated report execution test
Report generation (large / batch)	< 5 minutes	Manual execution with large dataset
Bulk data export	< 60 seconds for up to 100,000 rows	Automated export test
System availability (post-go-live)	> 99.5% during business hours	Monitoring dashboard review
Database backup completion window	Full backup within 2-hour window	SQL Server Agent job test
Cutover load duration	Full production load within maintenance window	Timed Dry Run 3 execution

4C.3 Performance Testing Activities

- Author performance test scripts covering the top 20 most-used application transactions (derived from Oracle Forms usage logs if available)
- Execute baseline single-user tests to establish minimum response times with no concurrent load
- Execute load tests at 25%, 50%, 75%, and 100% of expected peak concurrent user count
- Capture SQL Server wait stats, execution plans, and blocking reports during load test execution
- Identify and tune the top 10 highest-duration or highest-frequency queries : add indexes where required
- Re-execute load tests after tuning to confirm SLA thresholds are met
- Test SQL Server memory configuration under sustained load : verify no excessive paging
- Validate that SQL Server backup jobs complete within their scheduled windows even under business-hours load
- Document all tuning changes (indexes added, query plan changes) and apply to production deployment scripts

4C.4 Exit Criteria

- All SLA targets met at 100% concurrent user load
- Performance test results reviewed and approved by DBA and Development Lead
- All index additions and query tuning changes documented and version-controlled
- No unresolved blocking or deadlock conditions identified during testing
- Performance sign-off issued by DBA and Project Manager

PHASE 05

Training

End-User Enablement, System Administration & Support Readiness

5.1 Objectives

The Training phase ensures that all users, administrators, and support staff are proficient in the new .NET application before UAT commences and prior to production go-live. Effective training reduces resistance to change, minimizes post-go-live support calls, and accelerates adoption of the new system.

5.2 Audience Segments & Training Tracks

Audience	Track	Format	Duration
End Users	Core application navigation, daily workflows, data entry	Instructor-led + hands-on lab	4–8 hours
Power Users / Super Users	Advanced features, reporting, data exports, troubleshooting	Workshop (small group)	8–12 hours
System Administrators	SQL Server administration, user management, backups, monitoring	Technical deep-dive	8–16 hours
Help Desk / Support Staff	Common issues, error messages, escalation paths, known limitations	Reference guide + walkthrough	4 hours
Management / Executives	Dashboards, KPI reports, data insights in the new system	Demonstration session	1–2 hours

5.3 Training Deliverables

- User Manuals : role-based guides for each major functional area
- Quick Reference Cards : single-page laminated reference for common tasks
- Video Tutorials : recorded walkthroughs of key workflows for on-demand replay
- Training Environment : a dedicated non-production instance loaded with representative sample data
- FAQ Document : compiled from questions raised during training sessions
- Training Attendance Register : signed acknowledgment from all trained users
- Training Completion Report : submitted to Project Manager upon completion of all sessions

5.4 Training Environment Requirements

ENVIRONMENT	Training must use a dedicated environment seeded with anonymized/sanitized production-representative data. This environment must mirror the production .NET application and SQL Server version. Under no circumstances should training activities be conducted against a production or UAT database.

5.5 Super User Program

A network of Super Users (typically 1 per department) shall be identified and trained in advance of the broader end-user training. Super Users receive deeper training and serve as first-line support during and after go-live. They are also participants in the UAT phase.

5.6 Exit Criteria

- All identified users in scope have attended and acknowledged their respective training sessions
- Training Completion Report approved by Project Manager
- Super Users formally designated and confirmed ready for UAT support role
- Training materials finalized and published to the organizational knowledge base

PHASE 5B

Change Management & Communications

Stakeholder Engagement, Resistance Management & Go-Live Messaging

5B.1 Objectives

Technology migrations fail most often not because of technical errors, but because people are unprepared, uninformed, or resistant to the change. The Change Management and Communications phase runs concurrently with training and UAT and ensures that all affected stakeholders : from executives to end users : receive timely, accurate, and reassuring communications throughout the migration lifecycle.

5B.2 Stakeholder Communication Plan

Audience	Frequency	Channel	Key Messages
Executive / Sponsor	Bi-weekly	Status report + steering meeting	Budget, timeline, risk status, go/no-go readiness
Department Managers	Weekly	Email + team briefing	Impact on their team, training schedule, what changes for their users
End Users	At milestones	Email + town hall / all-hands	Why the change is happening, what to expect, when training occurs, who to contact
IT / Support Staff	Weekly	Team meeting + Slack/Teams	Technical details, deployment timeline, support procedure changes
Vendors / Integrations	As needed	Direct outreach	API endpoint changes, cutover window, re-testing schedule

5B.3 Change Management Activities

- Conduct a Change Impact Assessment : document what each role group's day-to-day work looks like before and after migration
- Identify change champions (Super Users) in each department to serve as advocates and first-line support
- Develop a resistance management plan : anticipate objections and prepare responses for the most common concerns
- Issue a Go-Live announcement to all staff no less than 2 weeks before the cutover window
- Communicate the maintenance window dates, expected downtime duration, and what users should do during the window
- Send a Post-Go-Live welcome communication on Day 1 with the new URL, login instructions, and help desk contact
- Conduct a 30-day post-go-live survey to capture user sentiment, adoption issues, and unmet needs

5B.4 Exit Criteria

- Change Impact Assessment completed and reviewed
- All scheduled stakeholder communications sent per plan
- Go-Live announcement issued to all affected users at least 2 weeks before cutover

- Help desk briefed and ready to receive post-go-live calls

PHASE 06

User Acceptance Testing (UAT)

Business Validation, Defect Resolution & Go-Live Authorization

6.1 Objectives

User Acceptance Testing is the formal process by which designated business users validate that the new .NET application and its SQL Server data meet all functional and non-functional requirements established during the project. UAT is the final quality gate before production go-live authorization is granted. It is conducted in a UAT environment loaded with Dry Run 3 data.

6.2 UAT Environment Specification

Application Version	Release Candidate build deployed to UAT environment
Database	SQL Server loaded with Dry Run 3 validated data
Data Scope	Full production-representative dataset (anonymized if required)
Access	UAT testers access via dedicated UAT URL / endpoint
Support	Developer and DBA on standby during UAT execution windows
Defect Tracking	All defects logged in the project defect tracking system with severity classification

6.3 UAT Test Scope : Functional Areas

- User Authentication and Role-Based Access Control
- All data entry forms previously served by Oracle Forms : field-level parity validation
- All business workflows: create, read, update, delete operations on all entity types
- Lookup / reference data validation : all code tables present and accurate
- Search and query functions : results match expected data from Oracle source
- Report outputs : .NET reports reconcile to legacy Oracle Reports on identical data sets
- Data export and import functions
- System notifications and alerts
- Business rules and validations enforced by the .NET application layer
- Historical data accessibility : all migrated legacy records browsable and correct
- Performance: page load and query response times within defined SLAs

6.4 Defect Severity Classification

Severity	Label	Definition	Resolution Requirement
SEV-1	**Critical / Showstopper**	System crash, data corruption, complete workflow failure, security breach	Must be resolved before UAT sign-off; go-live blocked
SEV-2	**Major**	Core business function broken, significant data discrepancy, no workaround available	Must be resolved before go-live

SEV-3	**Moderate**	Function impaired but workaround exists; minor data formatting issue	Must be resolved or risk-accepted before go-live
SEV-4	**Minor / Cosmetic**	UI cosmetic issue, non-critical label error, enhancement request	Logged for post-go-live sprint; does not block go-live

6.5 UAT Process Flow

11. UAT Kickoff Meeting : review scope, tester assignments, test scripts, and defect logging procedures
12. Test Script Execution : testers execute assigned test scripts and log pass/fail results
13. Defect Logging : all failures logged with steps to reproduce, screenshots, and severity classification
14. Defect Triage : daily triage meeting with development team to prioritize and assign defects
15. Defect Resolution & Retest : developers resolve defects; testers retest and verify closure
16. UAT Progress Reporting : daily status reports to Project Manager with open/closed defect counts
17. UAT Sign-off : when all SEV-1 and SEV-2 defects are closed and no open SEV-3 items are unresolved or risk-accepted, UAT Lead and Business Owner sign the UAT Acceptance Form
18. Go-Live Authorization : Project Sponsor issues formal go-live authorization based on UAT sign-off

6.6 UAT Exit Criteria

- 100% of test scripts executed
- Zero open SEV-1 (Critical) defects
- Zero open SEV-2 (Major) defects
- All SEV-3 defects resolved or formally risk-accepted by Business Owner
- UAT Acceptance Form signed by UAT Lead and Business Owner
- Go-Live Authorization issued by Project Sponsor

PHASE 6B

Parallel Run

Side-by-Side Output Validation : Oracle vs. .NET

6B.1 Objectives

The Parallel Run phase operates both the legacy Oracle Forms 6i system and the new .NET application simultaneously for a defined validation period. During this window, a selected set of business transactions are processed in both systems and their outputs are compared to confirm functional and data parity. The Parallel Run provides the highest level of confidence before the Oracle system is permanently decommissioned, and is the final safety net before the rollback window closes.

SCOPE Parallel Run is NOT intended to be a full production operation on both systems in perpetuity. It is a time-boxed validation exercise, typically 5 to 15 business days, executed immediately after go-live on a defined subset of transactions.

6B.2 Parallel Run Scope & Activities

- Define the parallel run transaction set: the top 20-30 most critical business transactions that must produce identical outputs
- Execute each in-scope transaction in the Oracle system (read-only/archive mode) and the .NET system independently
- Compare outputs field-by-field using a reconciliation script or a dedicated comparison report
- Log all discrepancies immediately into the parallel run defect log with severity classification
- Resolve all Critical and Major discrepancies within 24 hours : defects must be fixed in the .NET system; Oracle is not modified
- Re-execute failed comparisons after .NET fixes are deployed
- Produce a daily Parallel Run Parity Report showing pass/fail counts by transaction type

6B.3 Parallel Run Exit Criteria

Condition	Required Status
Transaction parity rate (all in-scope transactions)	**100% : all transactions produce matching outputs**
Open Critical discrepancies	**0 : all must be resolved before decommission approval**
Open Major discrepancies	**0 : all must be resolved before decommission approval**
Duration completed	**Minimum parallel run window has elapsed (as defined in project plan)**
Business Owner sign-off	**Written confirmation that outputs are acceptable and Oracle can be retired**

7. Cutover & Go-Live Strategy

Production cutover is executed during a pre-approved maintenance window, typically a weekend or public holiday, to minimize business impact. The cutover sequence follows the Dry Run 3 load procedure exactly, ensuring no surprises during the live event.

Cutover Checklist Summary

- Pre-Cutover: Freeze Oracle source system (read-only mode) : no new transactions
- Pre-Cutover: Final incremental data extract from Oracle 8 (delta since last dry run snapshot)
- T-0: Execute production SQL Server load using validated load scripts
- T+1hr: Execute full reconciliation report and compare to DR3 baseline
- T+2hr: Application team executes smoke test suite against production SQL Server
- T+3hr: Go/No-Go decision call with all stakeholders
- GO: DNS/URL switch to .NET application; Oracle Forms 6i decommissioned or placed in read-only archive mode
- NO-GO: Execute rollback procedure : revert DNS to Oracle Forms 6i; migrate team convenes post-mortem

ROLLBACK	Rollback Point of No Return (PNR): If the Go/No-Go decision has not been called within the first 50% of the maintenance window, rollback is automatically invoked. The Oracle system remains in read-only archive mode for a minimum of 90 days post-go-live.

PHASE 08

Hypercare & Post-Go-Live Stabilization

Intensive Support, Issue Triage & System Stabilization

8.1 Objectives

Hypercare is the intensive support period immediately following production go-live, typically lasting 30 days. During Hypercare, the full migration team remains available at elevated response levels to handle issues that were not surfaced during UAT or the Parallel Run. System monitoring is heightened, daily stand-up calls are mandatory, and all critical issues receive same-day response. Hypercare formally ends when the system achieves a defined stability threshold and normal BAU (Business As Usual) support takes over.

8.2 Hypercare Support Structure

Period	Support Level	Response SLA	Activities
Days 1–3	War room : all hands on deck	Critical: 30 min | Major: 2 hrs	Monitoring every 30 min; daily stand-up x2; all team on-call
Days 4–7	Elevated support	Critical: 1 hr | Major: 4 hrs	Monitoring hourly; daily stand-up; on-call rotation
Days 8–14	Enhanced support	Critical: 2 hrs | Major: 8 hrs	Monitoring 3x daily; daily stand-up; issue triage
Days 15–30	Transitional support	Critical: 4 hrs | Major: 24 hrs	Transition to BAU support team; knowledge transfer sessions

8.3 Hypercare Monitoring & Governance

- Daily Hypercare Status Report issued each morning: open issues, resolved issues, system metrics, user adoption stats
- SQL Server monitoring dashboards reviewed at defined intervals: CPU, memory, disk I/O, top wait stats, blocking queries
- Help desk ticket volume tracked daily : spike in volume signals a systemic issue requiring escalation
- All post-go-live defects logged with severity and resolved according to the Hypercare SLA table above
- Weekly Hypercare review meeting with Project Manager and Business Owner to assess stability trend
- Formal Hypercare Exit Review at Day 30: confirm all critical and major issues closed, BAU team briefed, monitoring handed over

8.4 Hypercare Exit Criteria

- Zero open Critical or Major issues in the defect log
- Help desk ticket volume trending at or below pre-migration baseline
- SQL Server performance metrics within SLA thresholds for 5 consecutive business days
- BAU support team has received all knowledge transfer documentation and is confirmed ready to operate independently

- Hypercare Exit Sign-off issued by Project Manager and Business Owner

PHASE 09

Legacy System Decommission

Oracle License Retirement, Data Archival & Infrastructure Teardown

9.1 Objectives

The Legacy System Decommission phase formally retires the Oracle Forms 6i application and Oracle 8 database, releases associated infrastructure, terminates Oracle license agreements, and ensures that all legacy data is properly archived in compliance with applicable data retention policies. Decommission cannot proceed until Parallel Run is complete and Hypercare has achieved its exit criteria.

HOLD PERIOD

The Oracle system must remain in read-only archive mode for a minimum of 90 days post-go-live before decommission activities begin. This provides a safety window in case previously unknown data or report references to the legacy system emerge during normal operations.

9.2 Decommission Activities

- Confirm Parallel Run and Hypercare exit criteria are met and documented : decommission is blocked until both are signed off
- Create a final point-in-time archive export of the Oracle 8 database using Oracle Data Pump : store in long-term archive storage
- Confirm the archive is readable and restorable : test by restoring to an isolated environment and running a row count check
- Document the archive storage location, encryption key custody, and retention period in the Data Retention Register
- Notify Oracle licensing contact of license retirement : obtain and file written confirmation of license termination
- Decommission Oracle Forms 6i application server: remove application, revoke service accounts, decommission VM or return hardware
- Decommission Oracle 8 database server: drop all schemas, remove Oracle software, decommission VM or return hardware
- Revoke all Oracle database user accounts and service accounts
- Update network diagrams, infrastructure documentation, and CMDB to reflect removal of Oracle components
- Notify downstream systems and integration partners that Oracle endpoints are permanently offline
- Archive all migration project artifacts (scripts, mapping documents, test results) to long-term project storage
- Conduct a migration retrospective: document lessons learned and recommended process improvements for future migrations

9.3 Data Retention Requirements

Data Category	Retention Period	Archive Format
Full Oracle 8 database archive (final export)	7 years minimum (or as defined by legal/compliance)	Oracle Data Pump .dmp file : encrypted at rest

Migration scripts and load logs	Duration of SQL Server system lifetime	Version control repository : compressed archive
Reconciliation and dry run reports	3 years minimum	PDF archive in project document store
UAT test results and sign-off documents	7 years minimum	PDF archive in project document store
Training materials	Duration of .NET application lifetime	Knowledge base / SharePoint : accessible to BAU team

9.4 Exit Criteria

- Oracle 8 database final archive confirmed readable and stored in approved long-term location
- Oracle license termination confirmed in writing
- All Oracle infrastructure decommissioned and removed from network diagrams and CMDB
- All migration project artifacts archived
- Lessons learned document completed and reviewed
- Decommission Sign-off issued by Project Manager and Infrastructure Lead

10. Roles & Responsibilities

Role	Responsibilities	Phase Involvement
Project Manager	Overall plan ownership, scheduling, risk management, stakeholder communication, go-live authorization	All phases
Migration Lead / ETL Architect	ETL design, transformation script development, dry run orchestration, reconciliation oversight	Extract, Transform, Clean, Load
Oracle DBA	Source system access, export scripts, schema documentation, legacy data profiling	Extract, Clean
SQL Server DBA	Target schema creation, load scripts, constraint management, performance tuning	Transform, Clean, Load, UAT
.NET Development Team	Application deployment to UAT/PROD, defect resolution, report validation	Load (DR2+), UAT
Business Analyst	Data mapping review, business rule documentation, UAT test script authoring	Transform, Clean, UAT
Training Coordinator	Training material development, session scheduling, attendance tracking	Training
UAT Lead	Test plan execution, tester coordination, defect triage, sign-off authority	UAT
Business Owner / Sponsor	Disposition decisions for cleansing, risk acceptance, final go-live authorization	Clean, UAT, Cutover

11. Risk Register

Risk	Likelihood	Impact	Mitigation
Oracle 8 export corruption or incomplete data extraction	Medium	Critical	Validate checksums; run extract twice and compare; DBA review
Data volume larger than estimated : extends dry run window	Medium	High	Benchmark load performance in DR1; optimize bulk insert strategy
PL/SQL to T-SQL conversion introduces logic errors	High	High	Parallel run of key business transactions on both systems during UAT
Table structure differences cause unmapped columns or lost data	High	Critical	Mandatory Data Mapping Specification with every column mapped before scripts are written
PK/FK key collisions from incorrect IDENTITY reseed	Medium	Critical	DBCC CHECKIDENT validation script mandatory after every dry run load
New FK constraints in SQL Server reject orphan Oracle records on load	High	High	Run orphan pre-check queries against Oracle source before DR1; remediate in Clean phase
Security role mapping grants excessive privileges	Medium	Critical	Privilege escalation review by IT Security before go-live; least-privilege principle enforced
Performance SLA failures under production load	Medium	High	Load test at 100% concurrent users in Phase 4C; index tuning before go-live
Key personnel unavailable during cutover	Low	High	Document runbook in detail; designate backup for each critical role
Legacy application still used during freeze window	Medium	Critical	Communicate blackout window 2 weeks in advance; enforce read-only lock
UAT defects volume exceeds resolution capacity	Medium	High	Pre-UAT code freeze; prioritize triage; negotiate SEV-3 risk acceptance
Parallel run discrepancies emerge after go-live	Medium	High	Maintain Oracle in read-only archive for 90 days; parallel run minimum 5 business days
User resistance leads to shadow Oracle usage post-go-live	Medium	Medium	Super User program; executive sponsorship; enforce Oracle read-only archive mode immediately
Oracle archive unreadable at decommission time	Low	High	Test archive restore to isolated environment before decommission sign-off

12. Migration Checklist

The following master checklist covers all phases of the complete J-Vo migration lifecycle. Each item must be formally signed off by the indicated approver before the migration may proceed to the subsequent phase. Items are grouped by phase for clarity.

#	Checklist Item	Approver	Status
PHASE 0 : Project Initiation & Discovery | Phase 0B : Environment Setup			
1	Project Charter signed by Project Sponsor	Project Sponsor	[ ]
2	Source & target system inventory documents complete and approved	PM + BA	[ ]
3	Migration scope document approved by Business Owner	Bus. Owner	[ ]
4	All environments provisioned: Dev, Staging, UAT, Training, Production	DBA + Infra	[ ]
5	Network connectivity and tooling readiness confirmed	DBA + Infra	[ ]
PHASE 1 : Extract | Phase 1B : Data Profiling			
6	Oracle 8 full export complete with row counts verified	DBA	[ ]
7	Extraction manifest approved and filed	Project Mgr	[ ]
8	Data Profile Report completed : all tables profiled across 8 dimensions	Migration Lead	[ ]
9	Anomaly Inventory reviewed; dispositions assigned by Business Owner	Bus. Owner	[ ]
PHASE 2 : Transform | Section 2A : Table Structure & Key Mapping | Phase 3 : Clean			
10	Data Mapping Specification approved (all tables, columns, PK/FK mappings)	BA + DBA	[ ]
11	PK reseed values calculated and IDENTITY_INSERT load scripts verified	DBA	[ ]
12	FK dependency load order documented; orphan pre-checks run on Oracle source	Migration Lead	[ ]
13	Transformation scripts unit tested against sample data	Migration Lead	[ ]
14	Data Quality Assessment Report issued	Migration Lead	[ ]
15	Business owner dispositions obtained for all cleaning exceptions	Bus. Owner	[ ]
16	Referential integrity checks pass 100% on cleansed data	DBA	[ ]
PHASE 4 : Load & Dry Runs | Phase 4B : Security | Phase 4C : Performance Testing			
17	Dry Run 1 complete : load scripts functional, schema created without errors	DBA + PM	[ ]
18	Dry Run 2 complete : reconciliation within 0.5% of source	DBA + Dev Lead	[ ]

19	Dry Run 3 complete : 100% reconciliation, load fits within cutover window	All Leads	[ ]
20	Key validation scripts pass 100%: PK, FK, orphan, IDENTITY reseed	DBA	[ ]
21	Security role mapping complete; all users provisioned; no privilege escalation	IT Security	[ ]
22	All SLA performance targets met at 100% concurrent user load	DBA + Dev Lead	[ ]
23	Rollback procedure tested and documented	DBA	[ ]
PHASE 5 : Training | Phase 5B : Change Management & Communications			
24	All user groups training complete : attendance records filed	Training Coord	[ ]
25	Super Users formally designated and confirmed ready for UAT support role	Project Mgr	[ ]
26	Go-Live announcement issued to all affected users (min. 2 weeks prior)	PM + Comms	[ ]
27	Help desk briefed and ready to receive post-go-live support calls	Support Lead	[ ]
PHASE 6 : UAT | Phase 6B : Parallel Run			
28	UAT test scripts 100% executed	UAT Lead	[ ]
29	Zero open SEV-1 and SEV-2 defects	UAT Lead	[ ]
30	UAT Acceptance Form signed by UAT Lead and Business Owner	Bus. Owner	[ ]
31	Go-Live Authorization issued by Project Sponsor	Project Sponsor	[ ]
32	Parallel Run complete : 100% transaction parity confirmed	Bus. Owner	[ ]
Cutover & Go-Live | Phase 8 : Hypercare | Phase 9 : Decommission			
33	Production cutover maintenance window approved	Project Sponsor	[ ]
34	Cutover Go/No-Go decision passed; .NET system live	All Leads	[ ]
35	Hypercare Exit Sign-off: zero critical/major issues, SLA met 5 days straight	PM + Bus. Owner	[ ]
36	Oracle final archive confirmed readable; license termination confirmed	DBA + PM	[ ]
37	All Oracle infrastructure decommissioned; CMDB updated; lessons learned filed	PM + Infra	[ ]

This document represents a comprehensive migration plan developed to showcase enterprise-level legacy system migration expertise.

J-Vo System Migration | Oracle Forms 6i + Oracle 8 → .NET + SQL Server | Portfolio Sample